

Comprehensive Market Analysis of Cloud Hosting Providers for zk-Proof Proving Projects

- AWS, GCP, and Azure dominate with broad GPU instance types, global regions, and mature automation support but face spot capacity issues.
- Specialized providers like CoreWeave, RunPod, and Lambda Labs offer competitive pricing, flexible spot markets, and zk-specific optimizations.
- Hetzner and OVHcloud provide cost-effective GPU servers with European focus but limited spot/preemptible options and regional coverage.
- Linode and DigitalOcean offer developer-friendly GPU instances with transparent pricing but limited spot capacity and smaller global footprint.
- Vast.ai's bidding marketplace provides lowest-cost GPU access but with variable reliability and no guaranteed uptime.

Executive Summary

For a zk-proof proving project requiring high-performance, cost-efficient, and flexible compute resources—particularly NVIDIA GPUs (A100, H100, L40, etc.), high-memory CPUs, and low-latency NVMe storage—the choice of cloud provider is critical. The analysis of major and specialized cloud hosting providers reveals a nuanced landscape:

- **AWS, GCP, and Azure** are the hyperscale leaders with extensive GPU instance portfolios, global data center coverage, and robust automation (Terraform/Ansible/API). They offer on-demand, spot/preemptible, and reserved pricing with discounts up to 90% on spot instances. However, they suffer from frequent spot capacity shortages and complex pricing that can lead to cost overruns. Their bare-metal and confidential computing options enhance security but add complexity.
- **CoreWeave** stands out for its focus on high-performance computing (HPC) and AI workloads, offering bare-metal GPU instances with InfiniBand interconnects and Kubernetes-native orchestration. It provides up to 60% reserved instance discounts and is used by zk projects for its scalable, secure, and high-bandwidth clusters. CoreWeave's infrastructure is optimized for distributed proving workloads but requires Kubernetes expertise.
- **RunPod** offers a developer-centric GPU cloud with containerized environments, root access, and transparent pricing starting at \$0.34/GPU-hour for RTX 4090 and \$1.99/GPU-hour for H100. It supports Terraform and integrates with SkyPilot, enabling rapid prototyping and distributed training. RunPod's spot market and community cloud provide cost-effective access but may face capacity constraints.



- **Lambda Labs** provides on-demand H100 GPUs at ~\$2.99/GPU-hour, targeting AI researchers and enterprises needing high-end hardware with hybrid cloud options. Its pricing is competitive with Azure but lacks spot/preemptible instances.
- **Vast.ai** operates a competitive bidding marketplace where GPU prices start around \$0.90/hour for H100, enabling cost savings but with variable host reliability and no uptime guarantees. Suitable for experimentation but not production workloads.
- **Hetzner** offers dedicated GPU servers in Europe starting at €3.49/month, with no spot/preemptible instances and limited GPU availability. Its low cost is attractive but lacks flexibility and global coverage.
- **OVHcloud** provides GPU instances optimized for AI and HPC workloads with European data center presence but limited spot capacity and outdated hardware in some regions.
- **Linode and DigitalOcean** focus on developer experience with straightforward pricing and GPU droplets starting at \$0.76/GPU-hour, but their spot capacity and global footprint are smaller compared to hyperscalers.

Comparative Tables

Table 1: GPU Instances Overview

Provider	Instance Name	GPU Model	vCPU	RAM (GB)	Storage Type/ Size	On-Demand Price		Spot Price (/hr)	Reserved Price (1Y/3Y)	Terraform Support	Custom Images Support
						Spot/Preemptible Price (/hr)	Discount %				
AWS	p4d.24xlarge	NVIDIA A100	96	1152	8× 1.9 TB NVMe SSD	32.77		19.66 (avg)	~80%	Yes	Yes
AWS	g5.48xlarge	NVIDIA L40S	192	768	4× 1.9 TB NVMe SSD	1.006		0.50 (avg)	~50%	Yes	Yes
GCP	a2-highgpu-8g	NVIDIA H100	96	768	8× 3.8 TB Local SSD	6.69		1.24 (preemptible)	~81%	Yes	Yes
Azure	NC H100 v5	NVIDIA H100	64	448	8× 1.9 TB NVMe SSD	6.98		12.0 (8-GPU ND H100 v5)	~80%	Yes	Yes
CoreWeave	N/A	NVIDIA H100	128	1024	NVMe (customizable)	6.16		N/A (reserved up to 60% off)	N/A	Yes	Yes
RunPod	RTX 4090	NVIDIA RTX 4090	16	64	NVMe	0.34		N/A (spot market)	N/A	Yes	Yes



Provider	Instance Name	GPU Model	vCPU	RAM (GB)	Storage Type/Size	On-Demand Price		Spot Discount %	Reserved Price (1Y/3Y)	Terraform Support	Custom Image Support
						(/hr)	Spot/Preemptible Price(/hr)				
RunPod	H100	NVIDIA H100	64	256	NVMe	1.99		N/A (spot market)	N/A	Yes	Yes
Lambda Labs	H100	NVIDIA H100	64	256	NVMe	2.99		N/A	N/A	Yes	Yes
Vast.ai	H100	NVIDIA H100	64	256	NVMe	0.90		N/A (bidding market)	N/A	No	No
Hetzner	GEX44	NVIDIA RTX 4000	14	64	2× 1.92 TB NVMe	0.0056 (€/hr)		N/A	N/A	No	No
Linode	GPU Linode	NVIDIA L40S	16	64	NVMe	0.0075 (min)		N/A	N/A	Yes	Yes
DigitalOcean	GPU Droplet H100	NVIDIA H100	64	256	NVMe	2.50		N/A	N/A	Yes	Yes

Table 2: High-RAM CPU Instances Overview

Provider	Instance Name	CPU Model	vCPU	RAM (GB)	Storage Type/Size	On-Demand Price		Spot Discount %	Reserved Price (1Y/3Y)	Terraform Support	Custom Image Support
						(/hr)	Spot/Preemptible Price(/hr)				
AWS	r6i.32xlarge	Intel Xeon	128	1024	EBS/NVMe	3.04		0.91 (avg)	~70%	Yes	Yes
GCP	m2-ultramem-128	Intel Xeon	128	1536	Local SSD	1.20		0.36 (preemptible)	~70%	Yes	Yes
Azure	E16s v5	Intel Xeon	16	128	Premium SSD	0.624		0.187 (spot)	~70%	Yes	Yes
CoreWeave	N/A	AMD EPYC	128	1024	NVMe (customizable)	0.50		N/A (reserved up to 60% off)	N/A	Yes	Yes
Linode	High Memory 128GB	Intel Xeon	16	128	NVMe	0.02		N/A	N/A	Yes	Yes



Provider	Instance Name	CPU Model	vCPU	RAM (GB)	Storage Type/Size	On-Demand Price		Spot Discount %	Reserved Price (1Y/3Y)	Terraform Support	Custom Image Support
						(/hr)	[Spot/Preemptible Price(/hr)]				
DigitalOcean	Memory Optimized	Intel Xeon	16	128	NVMe	0.03		N/A	N/A	Yes	Yes

Table 3: Provisioning & Automation Support

Provider	Terraform Support	Ansible Support	API Documentation Quality	Custom Image Support	Boot Time	Notes
AWS	Yes	Yes	Excellent	Yes	Fast	Full API, built-in Ansible plugins
GCP	Yes	Yes	Excellent	Yes	Fast	Full API, built-in Ansible plugins
Azure	Yes	Yes	Excellent	Yes	Fast	Full API, built-in Ansible plugins
CoreWeave	Yes	Limited	Good	Yes	Moderate	Kubernetes-native, requires expertise
RunPod	Yes	No	Good	Yes	Moderate	Integrates with SkyPilot, Terraform
Linode	Yes	No	Good	Yes	Moderate	REST API, Terraform modules
DigitalOcean	Yes	No	Good	Yes	Moderate	Simple API, Terraform modules
Hetzner	No	No	Limited	No	Slow	Limited automation support
OVHcloud	Yes	No	Moderate	Yes	Moderate	OpenStack/Kubernetes based

Table 4: Networking & Costs

Provider	Inter-Node Bandwidth	Latency	Data Egress Cost	Free Tier/ Egress	Notes
AWS	10 Gbps+	Low	Moderate	Limited	



Provider	Inter-Node Bandwidth	Latency	Data Egress Cost	Free Tier/ Egress	Notes
					Ultra-low-latency fiber, Direct Connect
GCP	200 Gbps	Low	Moderate	Limited	Global fiber network, Cloud Router
Azure	100 Gbps (InfiniBand)	Ultra-low	Moderate	Limited	ExpressRoute, InfiniBand networking
CoreWeave	InfiniBand	Ultra-low	Low	No	High-bandwidth cluster interconnects
RunPod	Moderate	Moderate	Low	No	Multi-region deployment
Linode	Moderate	Moderate	Moderate	No	Limited global footprint
DigitalOcean	Moderate	Moderate	Moderate	No	Limited global footprint
Hetzner	Moderate	Moderate	Moderate	No	European focus, limited global reach
OVHcloud	Moderate	Moderate	Moderate	No	European focus, limited global reach

Provider Deep Dives

AWS

AWS is the market leader with the broadest GPU instance portfolio and global infrastructure. It offers NVIDIA A100, L40S, and other GPUs with high RAM and NVMe storage options. AWS's spot instances provide up to 90% discounts but suffer from frequent capacity shortages, especially for high-demand GPU types. AWS supports Terraform and Ansible extensively, enabling automated provisioning and configuration. Its bare-metal and confidential computing options enhance security but add complexity. AWS's network is highly redundant with ultra-low-latency interconnects, ensuring high availability and performance. However, AWS's complex pricing and hidden costs can lead to budget overruns.

GCP

GCP offers NVIDIA H100 and other GPUs with up to 91% spot discounts and variable pricing. Its accelerator-optimized instances are designed for AI and HPC workloads, with high memory-to-vCPU ratios. GCP supports Terraform and Ansible well, enabling automated deployments. GCP's global fiber network ensures low-latency and high-bandwidth connectivity, ideal for distributed proving workloads. However, GCP's spot instance reliability varies, and its pricing can be unpredictable.



Azure

Azure provides NVIDIA H100 and other GPU instances with up to 90% spot discounts. Its NC and ND series VMs are optimized for AI and HPC workloads with InfiniBand networking for low latency. Azure supports Terraform and Ansible, enabling automated provisioning. Azure's network is designed for high-speed, low-latency connections, suitable for enterprise workloads. However, Azure's spot instances can be interrupted on short notice, and its pricing is complex.

CoreWeave

CoreWeave specializes in HPC and AI workloads, offering bare-metal GPU instances with InfiniBand interconnects and Kubernetes-native orchestration. It provides up to 60% reserved instance discounts and supports large multi-GPU clusters for distributed training. CoreWeave's infrastructure is optimized for zk-proving workloads requiring high bandwidth and low latency. CoreWeave requires Kubernetes expertise and is ideal for organizations training foundation models.

RunPod

RunPod offers containerized GPU environments with root access and transparent pricing starting at \$0.34/GPU-hour. It supports Terraform and integrates with SkyPilot, enabling rapid prototyping and distributed training. RunPod's spot market and community cloud provide cost-effective access but may face capacity constraints. RunPod is developer-friendly and suitable for fine-tuning large language models and distributed training.

Lambda Labs

Lambda Labs provides on-demand H100 GPUs at ~\$2.99/GPU-hour, targeting AI researchers and enterprises needing high-end hardware. Its hybrid cloud and colocation services allow scaling compute resources without losing control or performance. Lambda Labs is ideal for training large models and hybrid cloud deployments but lacks spot/preemptible instances.

Vast.ai

Vast.ai operates a competitive bidding marketplace for GPUs, offering prices starting around \$0.90/hour for H100. This enables cost savings but with variable host reliability and no uptime guarantees. Vast.ai is suitable for experimentation and cost-sensitive work but not for production workloads requiring guaranteed uptime.

Hetzner

Hetzner offers dedicated GPU servers in Europe starting at €3.49/month, with no spot/preemptible instances and limited GPU availability. Its low cost is attractive but lacks flexibility and global coverage. Hetzner's network spans 17 locations and 37 data centers, serving over 1.6 million customers in 140 countries.



OVHcloud

OVHcloud specializes in European cloud services, offering GPU instances optimized for AI and HPC workloads. Its network spans 17 locations and 37 data centers, serving over 1.6 million customers in 140 countries. OVHcloud's hardware is outdated in some regions, and its support is subpar compared to hyperscalers.

Linode

Linode offers GPU instances starting at \$0.0075/hour, designed for AI inference, big data processing, and video encoding. Linode supports Terraform and custom images but lacks spot/preemptible instances and has a smaller global footprint. Linode's GPU instances are optimized for high-throughput, low-latency inference at production scale.

DigitalOcean

DigitalOcean offers GPU Droplets starting at \$0.76/GPU-hour, designed for high-performance computing (HPC) workloads. DigitalOcean's GPU Droplets are equipped with NVIDIA Blackwell Ultra GPUs, delivering significantly expanded AI compute, memory capacity, and networking bandwidth. DigitalOcean supports Terraform and custom images but lacks spot/preemptible instances and has a smaller global footprint.

Recommendations

Cost-Optimized Picks

- **Vast.ai:** Best for experimentation and cost-sensitive workloads due to its competitive bidding marketplace offering the lowest GPU prices (~\$0.90/hr for H100). However, reliability varies by host, and it is not suitable for production workloads requiring guaranteed uptime.
- **RunPod:** Offers transparent pricing starting at \$0.34/GPU-hour and integrates with SkyPilot and Terraform, making it suitable for developers needing cost-effective GPU access with automation support.
- **DigitalOcean:** Provides straightforward pricing and GPU droplets starting at \$0.76/GPU-hour, ideal for developers valuing ease-of-use and predictable costs.

Performance Picks

- **CoreWeave:** Best for high-performance, low-latency, and distributed proving workloads requiring bare-metal GPU instances with InfiniBand interconnects and Kubernetes orchestration. CoreWeave's infrastructure is optimized for large-scale AI and ML workloads.
- **AWS, GCP, Azure:** Offer the broadest GPU instance portfolios, global data center coverage, and robust automation support. Their high-bandwidth, low-latency networks ensure high availability and performance for demanding workloads.



Automation-Friendly Picks

- **AWS, GCP, Azure:** Support Terraform and Ansible extensively, enabling automated provisioning, configuration, and orchestration. Their comprehensive APIs and built-in plugins facilitate seamless integration with existing workflows.
- **Linode, DigitalOcean:** Offer REST APIs and Terraform modules, suitable for developers needing automation but with smaller global footprints and limited spot capacity.

Avoid Unless Necessary

- **Hetzner, OVHcloud:** While cost-effective, they lack spot/preemptible instances and have limited global coverage and outdated hardware in some regions. Their support and managed services are subpar compared to hyperscalers.

Sources & Links

- AWS Pricing & Docs: [1](#) [2](#) [3](#) [4](#) [5](#) [6](#) [7](#) [8](#) [9](#) [10](#) [11](#) [12](#) [13](#)
- GCP Pricing & Docs: [14](#) [15](#) [16](#) [17](#) [18](#) [19](#) [20](#) [21](#) [22](#) [23](#) [24](#) [25](#) [26](#) [27](#) [28](#) [29](#)
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This comprehensive market analysis provides a detailed comparison of cloud hosting providers for zk-proof proving projects, focusing on GPU and high-memory CPU instances, pricing, automation support, networking, and geographic availability. The structured tables and deep dives enable informed decision-making based on project-specific requirements for performance, cost efficiency, and flexibility.

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- [1] [Amazon EC2 Pricing](#)
 - [2] [EC2 On-Demand Instance Pricing](#)
 - [3] [Amazon EC2 Spot Instances Pricing](#)
 - [4] [GPU Instance Pricing - AWS EC2 GPU Instances for AI/ML | DoiT Compute](#)
 - [5] [Selecting Ideal EC2 Instances for GPU Workloads on AWS](#)
 - [6] [AWS GPU Instance Pricing: P5, P4d, G5, Inf2 | Wring Blog](#)
 - [7] [AWS GPU Pricing Explained: Costs & Optimization Guide | TRG Datacenters](#)



- [8] [Accelerated computing Amazon EC2 instance types](#)
- [9] [Amazon EC2 G6 Instances | Amazon Web Services](#)
- [10] [Amazon EC2 G4 Instances — Amazon Web Services \(AWS\)](#)
- [11] [Amazon EC2 G5 Instances | Amazon Web Services](#)
- [12] [Virtual machine sizes overview - Azure Virtual Machines | Azure Docs](#)
- [13] [Azure VM Sizes & Pricing: A 2025 Guide for Engineering Teams | Sedai](#)
- [14] [About GPU instances | Compute Engine | Google Cloud Documentation](#)
- [15] [VM instance pricing - Compute Engine](#)
- [16] [GPU pricing | Google Cloud](#)
- [17] [Spot VMs | Compute Engine | Google Cloud Documentation](#)
- [18] [r/MachineLearning on Reddit: \[D\] Training on the cloud: GCP GPU pricing seems dramatically cheaper, why would you train on AWS or Azure?](#)
- [19] [H100 Rental Prices Compared: \\$1.49-\\$6.98/hr Across 15+ Cloud Providers \(2026\) | IntuitionLabs](#)
- [20] [Demystifying Google Cloud GPU Pricing: What You Need to Know - Oreate AI Blog](#)
- [21] [Pricing | Spot VMs | Google Cloud](#)
- [22] [GPU machine types | Compute Engine | Google Cloud Documentation](#)
- [23] [Accelerator-optimized machine family | Compute Engine | Google Cloud Documentation](#)
- [24] [Demystifying Google Cloud Platform Compute Engine Machine Types: A Comprehensive Guide | by Ayushmaan Srivastav | Medium](#)
- [25] [GCP Instance Types: Summary, Comparison & Recommendations](#)
- [26] [Machine families resource and comparison guide | Compute Engine | Google Cloud Documentation](#)
- [27] [Supported Cloud Instances | Clarifai Docs](#)
- [28] [GCP Instance Types Explained: Making the Right Choice for Your Workloads | CloudKeeper](#)
- [29] [Navigating GCP Instance Types: What To Use And When](#)
- [30] [Cloud GPU Pricing Comparison: AWS Vs Azure Vs GCP For AI Workloads \(2026\)](#)
- [31] [Azure Virtual Machine Pricing - GeeksforGeeks](#)
- [32] [Cloud GPU Pricing Comparison in 2025 — Blog](#)
- [33] [Microsoft Azure GPU Pricing 2026 – H100 & A100 Costs](#)
- [34] [Spot Virtual Machines – Spot Pricing and Features | Microsoft Azure](#)
- [35] [Azure cloud GPU server costs 2023 | Statista](#)
- [36] [Azure Pricing: The Complete Guide](#)
- [37] [Virtual machine sizes overview - Azure Virtual Machines | Microsoft Learn](#)
- [38] [Virtual Machine series | Microsoft Azure](#)
- [39] [Microsoft Azure Instance Types: Comprehensive Comparison](#)
- [40] [NV family VM size series - Azure Virtual Machines | Microsoft Learn](#)
- [41] [Microsoft Azure Instance Types: Detailed Explanation](#)
- [42] [Top 12 Cloud GPU Providers for AI and Machine Learning in 2026](#)
- [43] [H100 Rental Prices: A Cloud Cost Comparison \(Nov 2025\)](#)
- [44] [Cloud GPU Pricing Tracker — Compare AWS, Azure, GCP, CoreWeave, Lambda | Silicon Analysts](#)
- [45] [Cloud GPU Hosting Explained: Providers, Pricing, and Best ...](#)
- [46] [Hetzner Cloud for AI Projects — Complete GPU Server Setup & Cost Breakdown 2026 - DEV Community](#)



- [47] [Hetzner Cloud for AI Projects Complete GPU Server Setup & Cost Breakdown 2026 - by Dev To](#)
- [48] [Hetzner Cloud VPS Pricing Calculator \(Apr 2026\)](#)
- [49] [Hetzner presents a GPU server for trained AI models](#)
- [50] [GEX44](#)
- [51] [Hetzner | Review, Pricing & Alternatives](#)
- [52] [Cloud-hosting provider for developers & teams - Hetzner](#)
- [53] [High performance dedicated GPU server - Hetzner GEX131](#)
- [54] [Hetzner GPU Cloud | Compare & Launch with Shadeform](#)
- [55] [Hetzner GPU Server | Hacker News](#)
- [56] [How to Choose a Cloud GPU Provider for AI/ML Workloads in 2026 | DigitalOcean](#)
- [57] [Top 10 Microsoft Azure Alternatives for Cloud Apps in 2026 | DigitalOcean](#)
- [58] [Top 10 Cloud Service Providers in 2026 - Utho](#)
- [59] [Linode \(Akamai\) VPS: 75 Plans From \\$5.47/mo — Simple Cloud Hosting \(2026\)](#)
- [60] [Cloud Computing Pricing | Akamai - Linode](#)
- [61] [How does the GPU billing works? \(19777\) | Linode Questions](#)
- [62] [GPU Plans by Linode | VPSBenchmarks](#)
- [63] [Linode and NVIDIA Cloud GPU Instances | Akamai](#)
- [64] [Linode GPU instances offer accelerated computing power | TechTarget](#)
- [65] [Linode VM Pricing | Compare Instance Prices - Holori Calculator](#)
- [66] [Choose a compute plan](#)
- [67] [Linode Pricing - Actual Prices For All Plans, Enterprise Too](#)
- [68] [GPU Linodes](#)
- [69] [Top 10 Linode Alternatives for Cloud Computing in 2026 | DigitalOcean](#)
- [70] [GPU Droplets Pricing | DigitalOcean](#)
- [71] [Budget-Friendly Cloud Server Pricing | DigitalOcean](#)
- [72] [Droplet Pricing | DigitalOcean Documentation](#)
- [73] [Gradient™ AI GPU Droplets | DigitalOcean](#)
- [74] [7 Best Cloud GPU Platforms for AI, ML, and HPC in 2025 | DigitalOcean](#)
- [75] [DigitalOcean Pricing: VPS, GPU, Storage and More \(2026\)](#)
- [76] [r/digital_ocean on Reddit: Rent by the hour GPU option?](#)
- [77] [7 Strategies for Effective GPU Cost Optimization | DigitalOcean](#)
- [78] [Pricing | DigitalOcean](#)
- [79] [DigitalOcean GPU Droplets, scalable computing power on demand | DigitalOcean](#)
- [80] [DigitalOcean Servers Lineup at Cloudways | Cloudways Help Center](#)
- [81] [Ansible vs. Terraform](#)
- [82] [Terraform & Ansible: Unifying infrastructure provisioning and configuration management](#)
- [83] [Integrate Terraform with Ansible Automation Platform | HashiCorp Developer](#)
- [84] [Terraform + Ansible: When to Use Each and How to Integrate Them](#)

